Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-8331
Project Name	Team Enrollment Confirmation - Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Team Details

Team Size: 4

S.No	Name	Role	Team
1	SRIKAVI T	Team Leader	SWTID-2026-8331
2	ISWARYA S	Team Member	SWTID-2026-8331
3	YOGASRI S	Team Member	SWTID-2026-8331
4	KEERTHIKA L	Team Member	SWTID-2026-8331

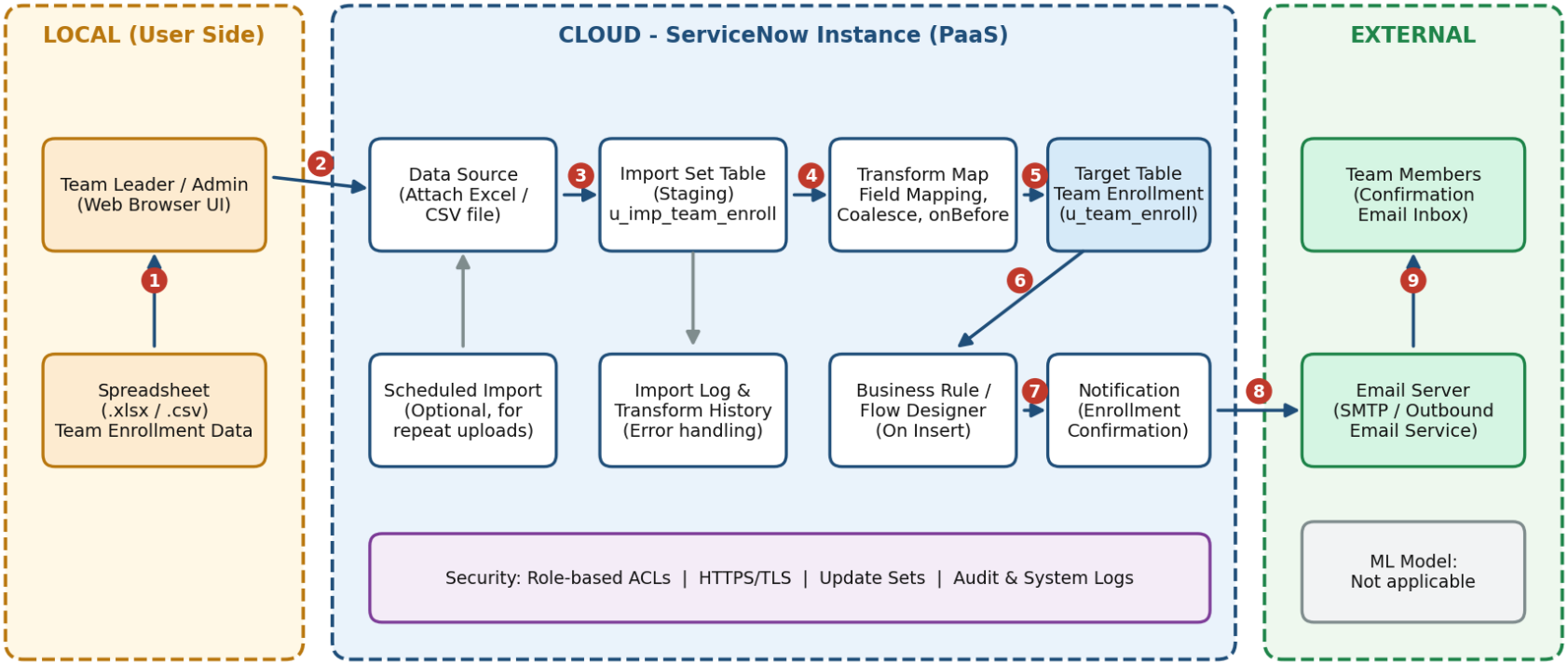
Project Overview

This project imports team enrollment data from a spreadsheet into ServiceNow using Import Sets and Transform Maps, stores it in a target table, and sends an enrollment confirmation to each team member. It replaces manual data entry with a repeatable, validated import.

Technical Architecture

Example reference: Team Enrollment Confirmation - Import Data using Transform Maps (Spreadsheet). The diagram below shows the local, cloud and external components, data storage and the flow between them.

Team Enrollment Confirmation - Import Data using Transform Maps: Technical Architecture



Numbered arrows = main process flow. Grey arrows = supporting flow (logging / optional scheduling).

Data Flow

1. The team leader / admin prepares the enrollment spreadsheet (.xlsx / .csv) locally.
2. The file is uploaded and attached to a Data Source in the ServiceNow instance.
3. Load Data copies the rows into the Import Set (staging) table.
4. The Transform Map maps columns to fields, coalesces on Team ID and validates values.
5. Valid rows are inserted or updated in the Team Enrollment target table; errors go to the import log.
6. A business rule / flow fires on insert and triggers the confirmation notification.
7. The email server delivers the confirmation to the team member's inbox.

Sample Field Mapping (Transform Map)

S.No	Spreadsheet Column (Source)	Target Field (u_team_enrollment)	Notes
1	Team ID	u_team_id	Coalesce field (unique key)
2	Member Name	u_member_name	Direct mapping
3	Role	u_role	Team Leader / Team Member
4	Email	u_email	Used for confirmation email
5	Enrollment Status	u_status	Set to 'Confirmed' by business rule

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web UI used by the team leader / admin to upload the enrollment spreadsheet, run the transform and view the imported records.	ServiceNow Web UI (Now Platform), Import Sets and Data Source forms; HTML, CSS, JavaScript (platform-provided)
2	Application Logic-1	Data import: the spreadsheet is attached to a Data Source and loaded into a staging (import set) table.	ServiceNow Import Sets, Data Source (Excel .xlsx / CSV)
3	Application Logic-2	Transform Map: maps spreadsheet columns to target fields, coalesces on a key field to avoid duplicates, and validates values before insert.	ServiceNow Transform Maps, field maps, onBefore transform script (server-side JavaScript)
4	Application Logic-3	Confirmation logic: when a new enrollment record is inserted, an event triggers the enrollment confirmation to the team member.	Business Rule / Flow Designer, ServiceNow Notifications & Events
5	Database	Staging table (import set) and target table holding team enrollment data (Team ID, Name, Role, Email, Status).	ServiceNow relational database (platform-managed tables)
6	Cloud Database	Database service hosted with the ServiceNow instance in the cloud.	ServiceNow Cloud instance database (Personal Developer Instance)
7	File Storage	Uploaded spreadsheet file is stored against the Data Source record.	ServiceNow Attachments (sys_attachment)
8	External API-1	Sends the enrollment confirmation email to team members.	SMTP / ServiceNow outbound email service
9	External API-2	Optional: lets other systems read enrollment records after import.	ServiceNow REST Table API (JSON) - optional
10	Machine Learning Model	Not applicable for this project.	N/A
11	Infrastructure (Server / Cloud)	Application deployed on the ServiceNow cloud instance; the user only needs a browser locally. Local Server Configuration: Windows / any OS, Chrome or Edge browser, MS Excel to prepare the file. Cloud Server Configuration: ServiceNow Personal Developer Instance (PDI), admin / import_admin role.	Local (browser) + Cloud (ServiceNow PaaS)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	ServiceNow is a proprietary platform, so no open-source framework is required. Open standards and formats are used for data exchange and scripting.	CSV, XLSX (Office Open XML), JSON / REST, SMTP, JavaScript
2	Security Implementations	Role-based access to import tables and target table, secured web access, validation of incoming data, and audit trail of imports.	ACLs and roles (admin / import_admin), HTTPS/TLS, transform validation, audit & system logs
3	Scalable Architecture	Layered design: presentation (UI), application (import / transform / notification logic) and data (tables). Staging table separates upload from processing so large files can be handled in batches.	3-tier on ServiceNow platform, Import Sets, scheduled imports
4	Availability	Application runs on the ServiceNow cloud, which handles hosting, redundancy and failover, so the import service stays available without local servers.	ServiceNow cloud infrastructure (platform-managed)
5	Performance	Transform runs on the server side in one pass; coalesce prevents duplicate records; notifications are queued as events so they do not slow the import.	Import set batch processing, coalesce, event queue

References

- https://c4model.com/
- https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
- https://www.ibm.com/cloud/architecture
- https://aws.amazon.com/architecture
- https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d